THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 6 — MATHS FULL SYLLABUS PAPER SET 08

Full Syllabus (Chapters 1–10) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The next number in the powers of 2 sequence $1, 2, 4, 8, 16, \ldots$ is: 1

(a) 32 **(b)** 48 **(c)** 36 **(d)** 24

2. The pattern 1, 4, 7, 10, … adds ___ each time: 1

(a) 3 **(b)** 1 **(c)** 4 **(d)** 2

3. A ray has: 1

(a) a fixed length
(b) two end points
(c) one starting point and extends endlessly in one direction
(d) no starting point

4. An angle of 250° is a: 1

(a) reflex angle **(b)** acute angle **(c)** obtuse angle **(d)** right angle

5. Which one is a palindrome? 1

(a) 1223 **(b)** 2332 **(c)** 2323 **(d)** 2331

6. The Kaprekar routine uses: 1

(a) the largest and smallest numbers formed from the digits, then subtraction
(b) division by 7
(c) addition of all digits
(d) multiplication by 9

7. In a pictograph, half a symbol is used to show: 1

(a) double the value **(b)** nothing
(c) half of the key value **(d)** the title

8. In a pictograph where one symbol stands for 10, how many symbols are needed for 60? 1

(a) 7 symbols **(b)** 8 symbols **(c)** 5 symbols **(d)** 6 symbols

9. The prime factorisation of 12 is: 1

(a) 2 × 3 × 3 **(b)** 2 × 2 × 3 **(c)** 2 × 6 **(d)** 3 × 4

10. A number is divisible by 4 if its last two digits form a number divisible by: 1

(a) 2 **(b)** 4 **(c)** 3 **(d)** 5

11. 1 m² equals: 1

(a) 10,000 cm² **(b)** 1,000 cm² **(c)** 100 cm² **(d)** 100,000 cm²

12. The area of a rectangle is: 1

(a) length × breadth **(b)** length + breadth
(c) half of length × breadth **(d)** 2 × (length + breadth)

13. Which is greater: $\frac{7}{10}$ or $\frac{2}{3}$? 1

(a) cannot be compared **(b)** $\frac{2}{3}$
(c) both are equal **(d)** $\frac{7}{10}$

14. $\frac{5}{6} + \frac{1}{3}$ equals: 1

(a) $\frac{67}{60}$ **(b)** $\frac{7}{6}$ **(c)** $\frac{6}{9}$ **(d)** $\frac{5}{4}$

15. Number of diagonals of a rectangle: 1

(a) 4 **(b)** 2 **(c)** 3 **(d)** 1

16. A rectangle of sides 14 cm and 9 cm has perimeter: 1

(a) 23 cm **(b)** 48 cm **(c)** 46 cm **(d)** 126 cm

17. A figure that matches itself only after a full turn has an order of rotational symmetry of: 1

(a) 360 **(b)** 2 **(c)** 1 **(d)** 0

18. The angle of rotational symmetry of an equilateral triangle is: 1

(a) 90° **(b)** 60° **(c)** 180° **(d)** 120°

19. The value of $8 + (-3)$ is: 1

(a) 7 **(b)** 5 **(c)** 3 **(d)** −5

20. If the temperature rises from −4°C by 7°C it becomes: 1

(a) −3°C **(b)** −11°C **(c)** 11°C **(d)** 3°C

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 10–15 words each.

21. How many dots are there in the 6th triangular arrangement? 2

22. How many days are there from 12 March to 12 April? 2

23. Data: 5, 3, 5, 2, 3, 5. What is the frequency of 5? 2

24. List the prime numbers between 20 and 40. 2

25. Find the perimeter of a rectangle of sides 12 cm and 7 cm. 2

26. What fraction of an hour is 15 minutes? 2

27. What is meant by ‘equidistant’? 2

28. What is the absolute value of −15 and of 15? 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. What is the connection between the number of sides of a regular polygon and the counting numbers? 3

30. Arrange the angles in increasing order: 135°, 20°, 90°, 270°, 180°. 3

31. How do artistic and aesthetic considerations help in presenting a graph? 3

32. Find the area of a square whose perimeter is 36 cm. 3

33. Using the properties of diagonals, how can you check if a quadrilateral is a rectangle? 3

34. List the quadrilaterals with 2 lines of symmetry. 3

35. Are 35 and 64 co-prime? Find their HCF. 3

OR

Three bells ring every 6, 8 and 12 minutes. After how many minutes will they ring together? 3

36. A shopkeeper has a profit of ₹500 on Monday and a loss of ₹800 on Tuesday. What is the overall result? 3

OR

A lift goes down 4 floors then up 7 floors from floor 2. At which floor does it stop? 3

SECTION — D (Long Answer Type) 4 × 5 = 20

Answer in about 80–100 words each.

37. **Case study:** A student turns a rotating arm through 45°, 90°, 135° and 180° in turn. 5

(i) Name the types of the angles.
(ii) What fraction of a full turn is 135°?
(iii) How many steps of 45° complete a full turn?
(iv) Which angle is a right angle?

OR

(i) Draw angles of 40°, 90° and 150°. 5
(ii) Classify them.
(iii) Arrange them in descending order.

38. (i) Write numbers between 1,000 and 2,000 whose digits sum to 3. 5
(ii) How many?
(iii) Which is the smallest?

OR

(i) Why is the Collatz conjecture famous? 5
(ii) Test it for 13.
(iii) What did you observe?

39. **Case study:** A class of 40: $\frac{3}{8}$ walk, $\frac{1}{4}$ cycle, rest use a bus. 5

(i) How many walk?
(ii) How many cycle?
(iii) Fraction using a bus?
(iv) How many use the bus?

OR

(i) State why the denominator cannot be zero. 5
(ii) Write the smallest unit fraction with denominator ≤ 10.
(iii) Order $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{10}$.

40. (i) Explain the symmetry of a regular polygon with n sides. 5
(ii) Apply for n = 8.

OR

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Case study: A tailor cuts a paper folded in half to create a heart. 5

(i) What symmetry does the cut-out have?

(ii) How many lines?

(iii) Does it have rotational symmetry?

(iv) Give another example.

END OF QUESTION PAPER **Total: 80 Marks**

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